

Salem Werdyani, Ph.D.

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Professional Summary

Distinguished scientist in human genetics, bioinformatics, AI-based research analyses, and statistical genomics. Expert in machine and deep learning approaches to unravel the genetic architecture of complex diseases, using large-scale GWAS, WES, WGS, single-cell RNA-seq, multi-omics data for patient stratification, drug discovery, and precision medicine. Author of 10+ peer-reviewed papers with an established record of competitive funding, cross functional team leadership, and mentorship.

Education

Ph.D. in Medicine (Human Genetics), Memorial University of Newfoundland (2018–2023).

M.Sc. in Medicine (Human Genetics), Memorial University of Newfoundland (2012–2016).

Post-Graduate Certificate in Bioinformatics, Centennial College (2010–2011).

B.Sc. Biology, UAE University (1993–1998).

Additional Training

Artificial Intelligence and Data Science for Leaders | The University of Chicago (May 2026).

Management and leadership (Becoming a Leader) program | The University of Chicago (March 2026).

The Longitudinal Data Analysis Using R | Statistical Horizons (Oct 2020).

The Human and Mammalian Genetics and Genomics | The 59th McKusick Short Course, the Jackson Laboratory, Bar Harbor, Maine. USA (July 2018).

Research Experience

Lead scientist | IPPH | The University of Chicago (MAY 2026 - Present)

- Lead interdisciplinary research programs in human genetics, bioinformatics, statistical genomics, and multiomics to advance understanding of complex diseases and therapeutic responses.
- Design scalable computational workflows and apply ML/statistical modeling to large-scale biomedical datasets to support precision medicine initiatives.
- Establish and maintain collaborations with academic institutions, healthcare organizations, and industry partners. Secure competitive research funding through grant development. Disseminate findings through high-impact peer reviewed publications and international conferences.

Postdoctoral Fellow | Khan Lab | Faculty of Medicine | The University of Chicago (2024 - Present)

- Led research on germline genetic influences on CAR T-cell therapy outcomes in B-cell hematological malignancies patients; applied PRS, fine-mapping (SuSiE/FINEMAP/cTWAS), and single-cell analyses.
- Discovered a novel rare T-cell subtype implicated in CAR T-cell therapy resistance via Leiden clustering and trajectory inference on 100K+ single cells.

Ph.D. Researcher | Zhai Lab, Memorial University of Newfoundland (2018 - 2023)

- Used GWAS/WGS/WES and metabolomic data to identify genetic and metabolic endotypes and biomarkers in osteoarthritis progression and joint replacement outcomes.
- Identified genetics and metabolomic markers linked to muscle strength reduction over 10 years.
- Published multiple high-impact peer-reviewed papers and presented findings at national and international conferences including Canadian Orthopedic Association meetings, OARSI, and ASHG meetings.

M.Sc. Researcher | Savas Lab, Memorial University of Newfoundland (2012 – 2016)

- Conducted computational prediction and analysis of CNV profiles in colorectal cancer patients using genome wide SNP genotyping datasets.
- Implemented rigorous statistical models and bioinformatics QC pipelines to improve CNV detection accuracy.
- Mapped genes and disrupted pathways associated with predicted CNVs to assess their biological implications in colorectal cancer.

Selected Publications

1. **Salem Werdyani**, Dawn Aitken, Zhiwei Gao, Ming Liu, Edward W. Randell, Proton Rahman, Graeme Jones, Guangju Zhai. *Metabolomic signatures for the longitudinal reduction of muscle strength over 10 years*. Under review in the BMC Skeletal Muscles Journal. PMID: 35130970, DOI: 10.1186/s13395-022-00286-9.
2. **Salem Werdyani**, Ming Liu, Hongwei Zhang, Guang Sun, Andrew Furey, Edward W. Randell, Proton Rahman, Guangju Zhai. *Endotypes of primary osteoarthritis identified by plasma metabolomics analysis*. Rheumatology (Oxford), 2020. PMID: 33159799, DOI: 10.1093/rheumatology/keaa693.
3. **Salem Werdyani**, Yajun Yu, Georgia Skardasi, Jingxiong Xu, Konstantin Shestopalok, Wei Xu, Elizabeth Dicks, Jane Green, Patrick Parfrey, Yildiz E. Yilmaz, and Sevtap Savas. *Germline INDELs and CNVs in a cohort of colorectal cancer patients: their characteristics, associations with relapse-free survival time, and potential time-varying effects on the risk of relapse*. Cancer Medicine, 2017. PMID: 28544645, DOI: 10.1002/cam4.1074.

Selected Awards & Scholarships

- Top Canadian Researchers Award, Canadian Orthopedic Association Annual Meeting (2022).
- Canadian Arthritis Society Top 10 Research Advances of 2021.
- Canadian Arthritis Society PhD Salary Award for 3 years (2019–2022).
- The Terry Fox Research Institute (TFRI) and Beatrice Hunter Cancer Research Institute travel bursary (July 2015).
- The Canadian Institutes of Health Research (CIHR) Travel Award (Spring 2015).

Technical Expertise

Languages/tools: R, Python, Bash, SQL, Git, Docker, Nextflow, Snakemake. **ML/AI:** TensorFlow, JAX, scikit-learn, XGBoost, Auto-sklearn. **Genomics:** GWAS, WES/WGS, RNA-seq, scRNA-seq, ChIP-seq, PRS (LDpred2, PRS-CS, Lassosum2), fine-mapping (SuSiE, FINEMAP). **Variant tools:** PLINK, GATK, ANNOVAR, SnpEff. **Cloud/infra:** Google Cloud Platform, Digital Research Alliance of Canada.

Teaching, Mentorship & Leadership

- Delivered lectures for undergraduate, graduate, and medical courses in human molecular genetics, population genetics, and genetic epidemiology.
- Led multidisciplinary research teams; mentored undergraduate/graduate students and clinical residents in research design and career development.